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Boltzmann Equation Solver for Thermalization

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Based on arXiv:2603.28848
Comput. Phys. Commun. 327 (2026) 110295
Code: github.com/best-hep/best

Institut Pascal, Orsay
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Outline

Part I — Reaching equilibrium

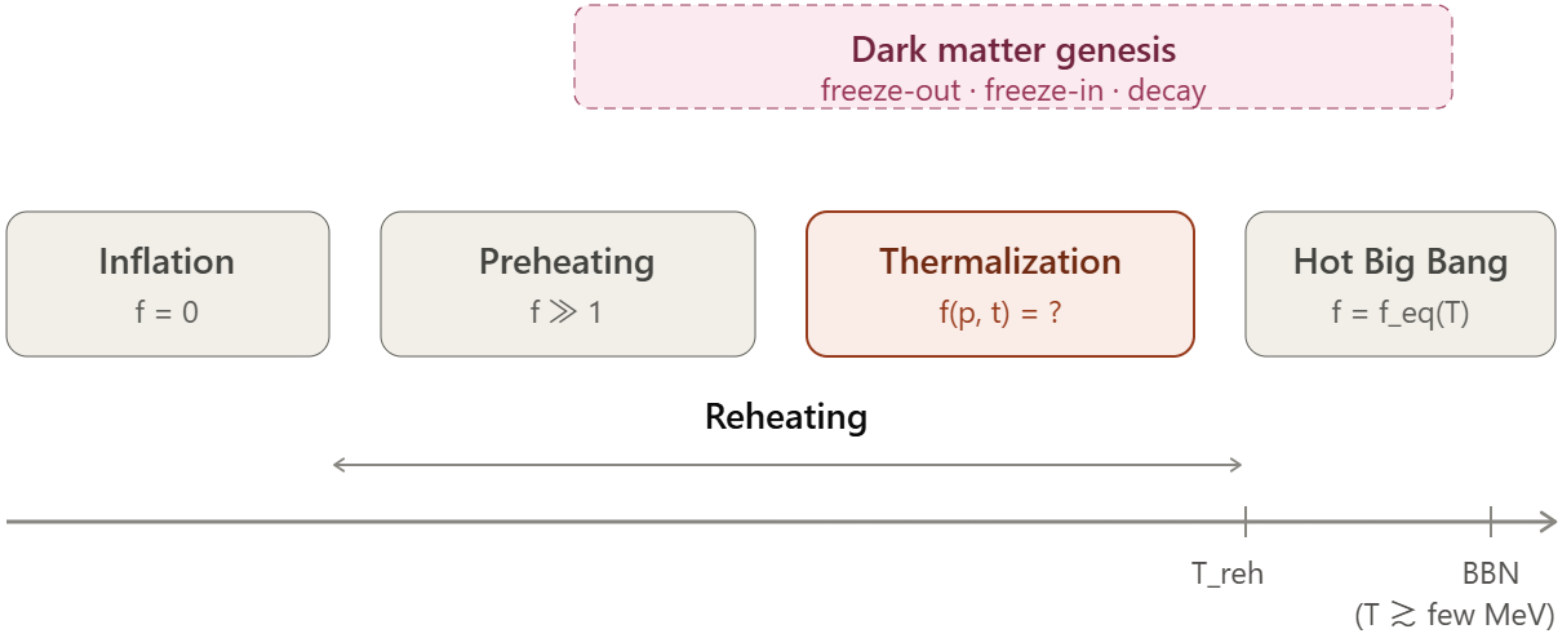
- Motivation: when do we need the full Boltzmann equation?
- Direct Monte Carlo for arbitrary n → m processes
- Validation & thermalization; what changed since the paper

Part II — Leaving equilibrium

- From flat spacetime to the expanding universe
- Sub-threshold freeze-out (aka forbidden DM): relic abundance from the full f(p, t)
- Both at once: a stress test

Outlook & Summary

Particle production in the early universe



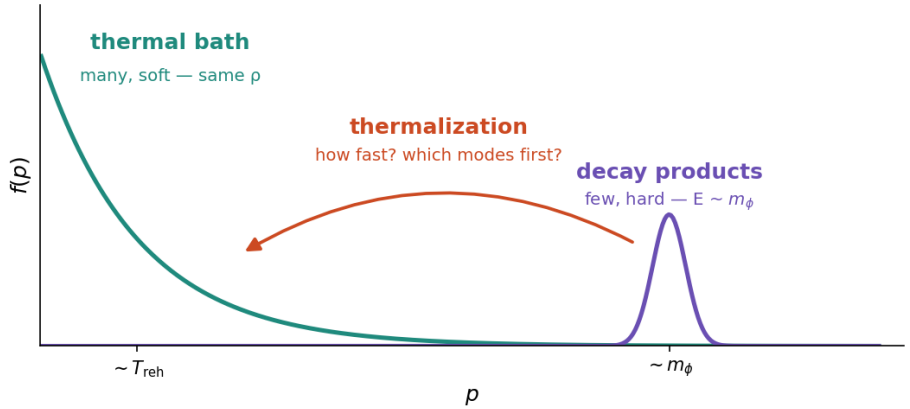
The reheating temperature

Unmeasured: anywhere between ~MeV (BBN) and ~10^{15} GeV, and it decides which dark-matter and baryogenesis scenarios are even possible.

The standard estimate of the reheating temperature:

$$T_{\rm reh} \sim \left(\frac{90}{\pi^2 g_*}\right)^{1/4} \sqrt{\Gamma_\phi M_{\rm Pl}}$$

assumes instantaneous thermalization of the inflaton decay products.



Thermalization (reaching full equilibrium) is governed by the Boltzmann equation with number-changing processes.

Dark matter genesis

The second episode on the timeline: a relic abundance, fixed by how χ falls out of equilibrium.

Where the standard machinery breaks:

- **Cannibal / SIMP** — relic set by 3 → 2 self-interaction**s**
- **Freeze-in → freeze-out transition** — ⟨σv⟩ averaging invalid
- **Dark-sector internal thermalization** — T_χ must emerge
- **Sub-threshold annihilation** — only particles above threshold can annihilate (→ Part II)

Different settings, one common demand: the **shape of f(p, t)**.

What existing tools assume

Most existing approaches assume the shape of f — the only question is how much:

Level	Solves for	Assumes	Tools
nBE	n(t)	equilibrium shape of f ; $\langle\sigma v\rangle$	micrOMEGAs, MadDM, DarkSUSY, DRAKE
cBE	n(t), T(t)	kinetic equilibrium (shape up to T, μ)	DarkSUSY, DRAKE
relaxation time	f (p, t)	$C[f] \approx -(f - f_{eq})/\tau$: one timescale, no momentum-exchange information	—
fBE	f (p, t)	nothing about the shape	DRAKE (2 → 2) · BEST (any n → m)

The collision integral and its cost

$$\frac{\partial f_a}{\partial t} - Hp\frac{\partial f_a}{\partial p} = \sum_{\text{processes}} C_a[f]$$

$$\boxed{\frac{df_a(q,t)}{dt} = \sum_{\text{processes}} C_a[f]}$$

'Boltzmann equation'

$$C^{(k)}(\mathbf{p}) = \frac{1}{2E}\int d\Pi\, |\mathcal{M}|^2\, \Lambda$$

$$\Lambda = \prod_{i\in\text{in}} f_i \prod_{j\in\text{out}} (1 \pm f_j) - \prod_{j\in\text{out}} f_j \prod_{i\in\text{in}} (1 \pm f_i)$$

$$d\Pi = (2\pi)^4 \delta^{(4)}\left(\sum_{i\in\text{in}} p_i - \sum_{j\in\text{out}} p_j\right) \prod_{l\neq k} \frac{d^3 p_l}{(2\pi)^3 2E_l}$$

- The right-hand side is an integral over every other particle — 3(n_total − 1) dimensions (2→2: 9, 2→3: 12) — with a δ^4 inside — at every grid point, every time step. For a general $|M|^2$, no analytical reduction exists beyond $2 \to 2$.

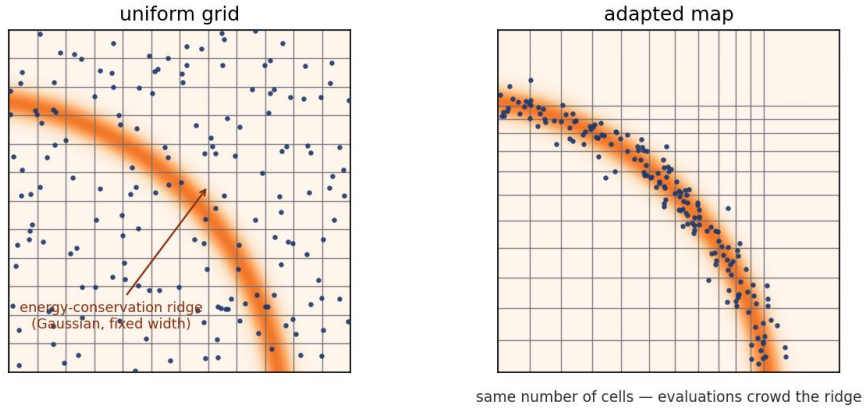
- Can we just integrate it directly?

Handling the delta function (<v1.3)

- In high dimensions there is one candidate: Monte Carlo, error ∝ 1/√N_eval. But the integrand contains δ^4: random samples never hit a zero-measure surface.

- Momentum (3 δ's): eliminated exactly. The conserved particle is determined, not sampled, and 3 dimensions are gone: 3(n_total − 2) (2→2: 6, 2→3: 9). Roles: observed (pinned at p) · conserved · integrated.

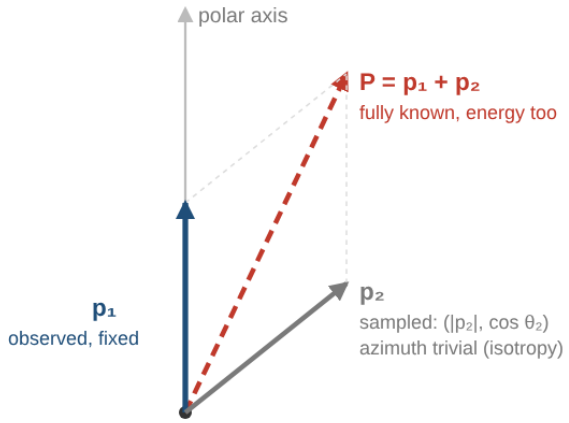
- Energy (1 δ): no variable left to solve for → narrow Gaussian



Vegas (Lepage '78; '21)

Exact kinematics: the last pair as a two-body decay

1. Lab frame: sample p_2 only

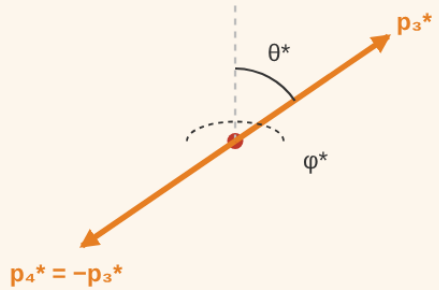


- 2 random numbers so far
- p_3, p_4 are **not** sampled as free vectors
- all we know: $p_3 + p_4$ must equal P
- $s = P^2 < (m_3 + m_4)^2$ → this sample gets weight 0

2. Rest frame of P: two-body decay

example 2→2: 1 + 2 → 3 + 4, particle 1 observed

a "fake particle" of mass √s at rest
(nothing real — just the 3+4 system seen from P)



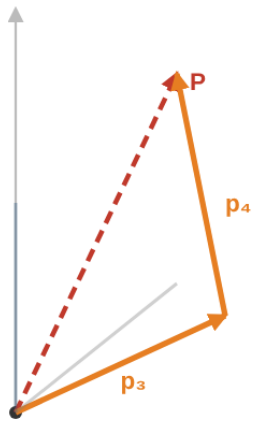
$|p^*| = \lambda^{1/2}(s, m_3^2, m_4^2) / (2\sqrt{s})$ — **fixed by s, not sampled**

$E_3^* = (s + m_3^2 - m_4^2) / (2\sqrt{s})$

sampled: direction $(\cos\theta^*, \varphi^*)$ only → **2 numbers, two legs**

δ^3 and $\delta(E)$ both used up here;
Jacobian = two-body phase space $p^*/(4\pi\sqrt{s})$

3. Boost back



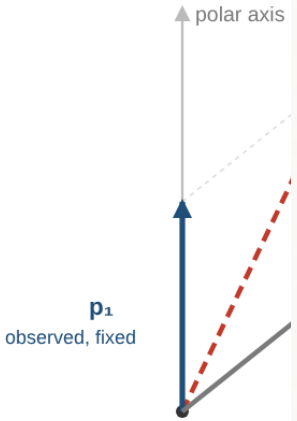
- boost p_3^*, p_4^* with $\Lambda(P)$ into the lab frame
- $p_3 + p_4 = P$ **by construction** — nothing to check, no Gaussian, no width parameter
- total: 2 + 2 = **4 numbers** for the whole 2→2 sample

general n→m: every free leg but the last two is sampled as in panel 1 (3 numbers each; the first azimuth is trivial), the last two as in panel 2

dimensions = **3 n_total − 8** · 2→2: 4 · 2→3: 7 · 1→2: 1 (real parent, no free leg, one angle trivial)

Exact kinematics: the last pair as a two-body decay

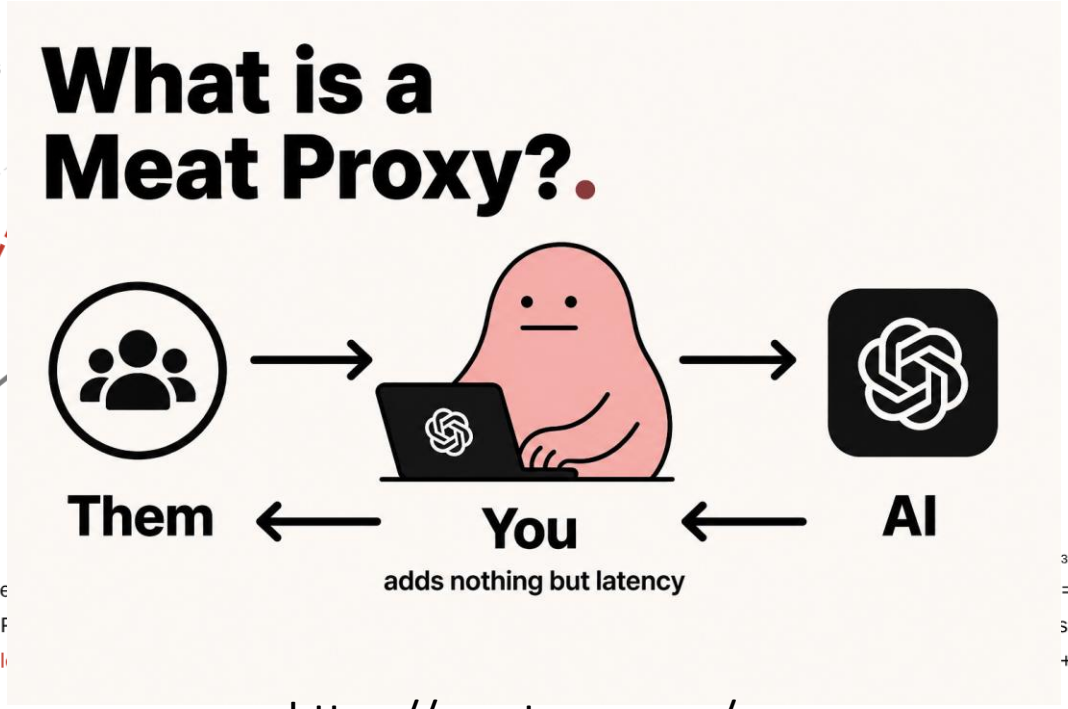
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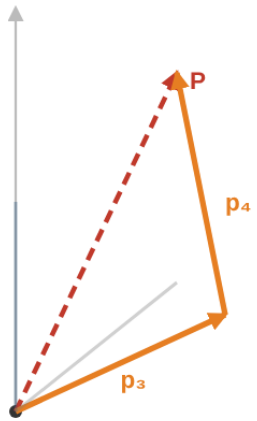
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https://meatproxy.me/

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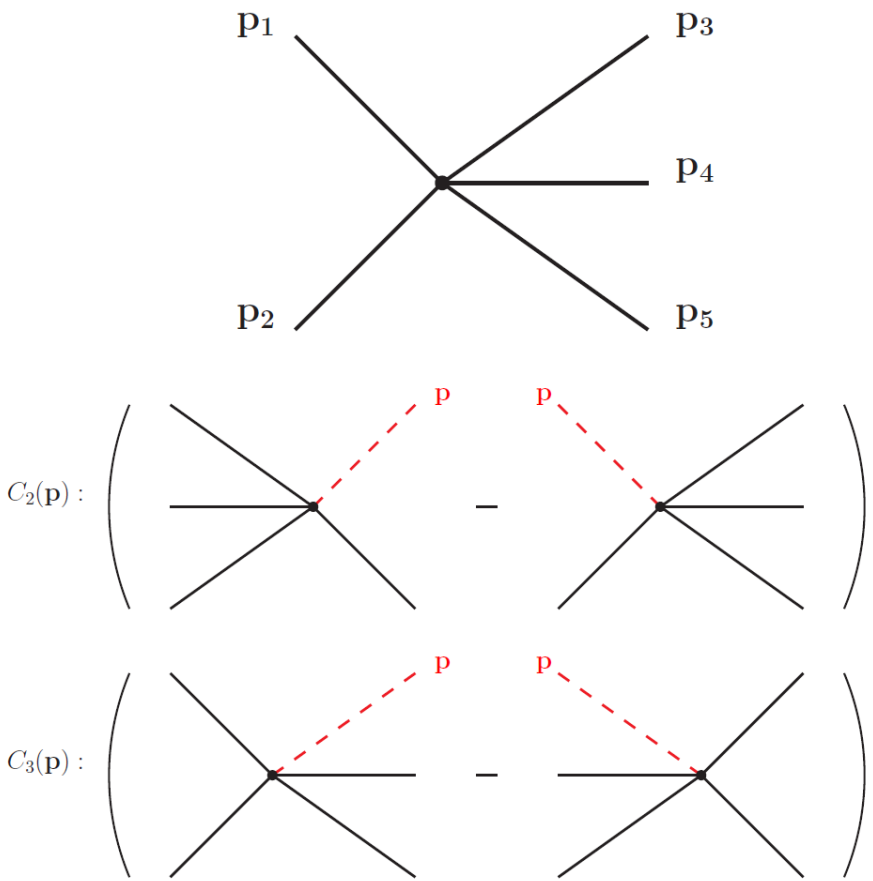


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The identical-particle decomposition



One particle is pinned — but which one, when they are identical?

For ϕϕ ↔ ϕϕϕ, p can sit on the 2-side or the 3-side

C2(p) — p on the 2-side:

gain: f3f4f5 (1+f (p))(1+f2)

loss: f (p)f2 (1+f3)(1+f4)(1+f5)

C3(p) — p on the 3-side:

gain: f1f2 (1+f (p))(1+f4)(1+f5)

loss: f (p)f4f5 (1+f1)(1+f2)

Two different integrals ⇒ keep both, slot-weighted: $C_\phi = 2\,C_2 + 3\,C_3$

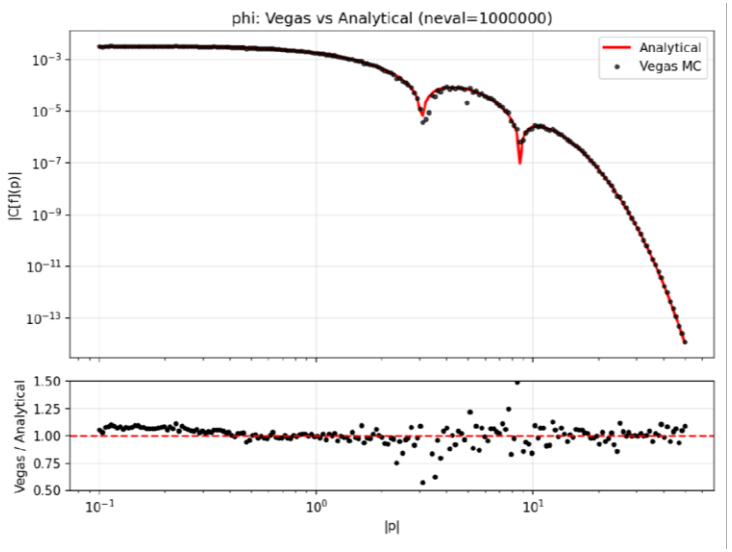
Validation: semi-analytical 2 → 2 benchmark

Independent cross-check, following Ala-Mattinen et al. ’22:

$$C_{\mathrm{BW}}(p_1) = \frac{2}{(2\pi)^4}\frac{1}{2E_1}\int dp_3\,\frac{p_3^2}{2E_3}\int dp_4\,\frac{p_4^2}{2E_4}\,F(p_1,p_3,p_4)\,f_3 f_4(1\pm f_1)(1\pm f_2)$$

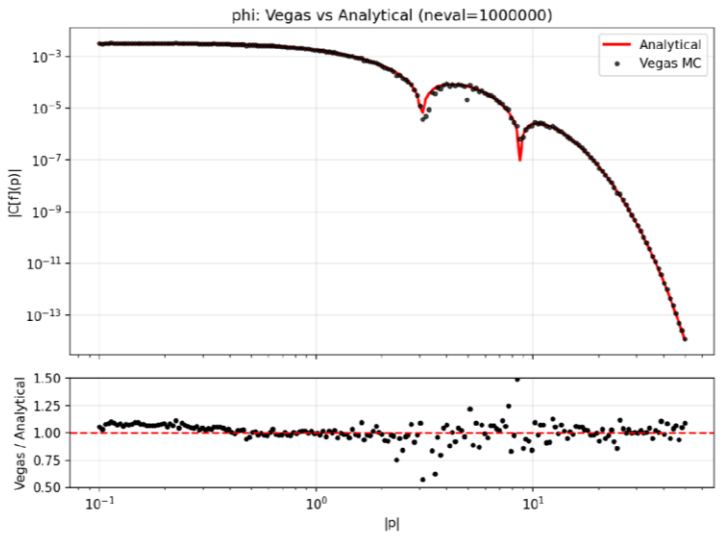
- angular integrations done analytically; 9 → 2 dimensions
- energy conservation exact (E2 = E3 + E4 − E1, no Gaussian)
- ⇒ tests both the MC accuracy and the systematic error from the finite δ-width

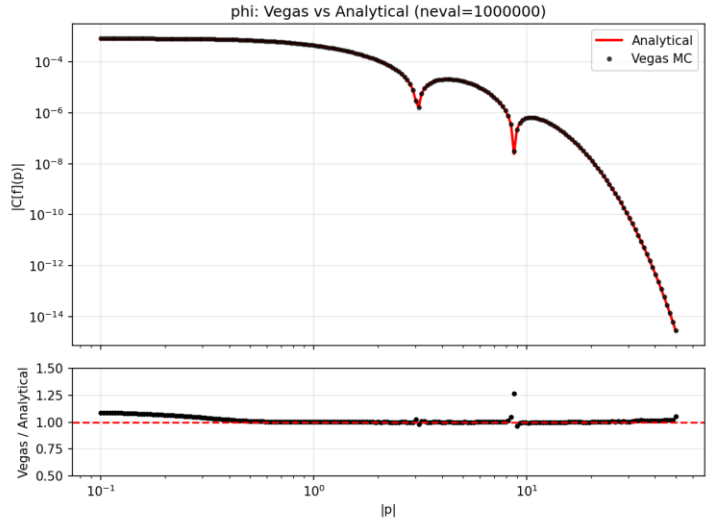
Benchmark result



- What is compared: the collision integral itself, Cf — Vegas vs. the exact semi-analytical result
- Lower panels: the net rate (gain − loss) — ratio deviations near its zero crossings
- Agreement at the few-% level wherever the rate is sizable.

Benchmark result

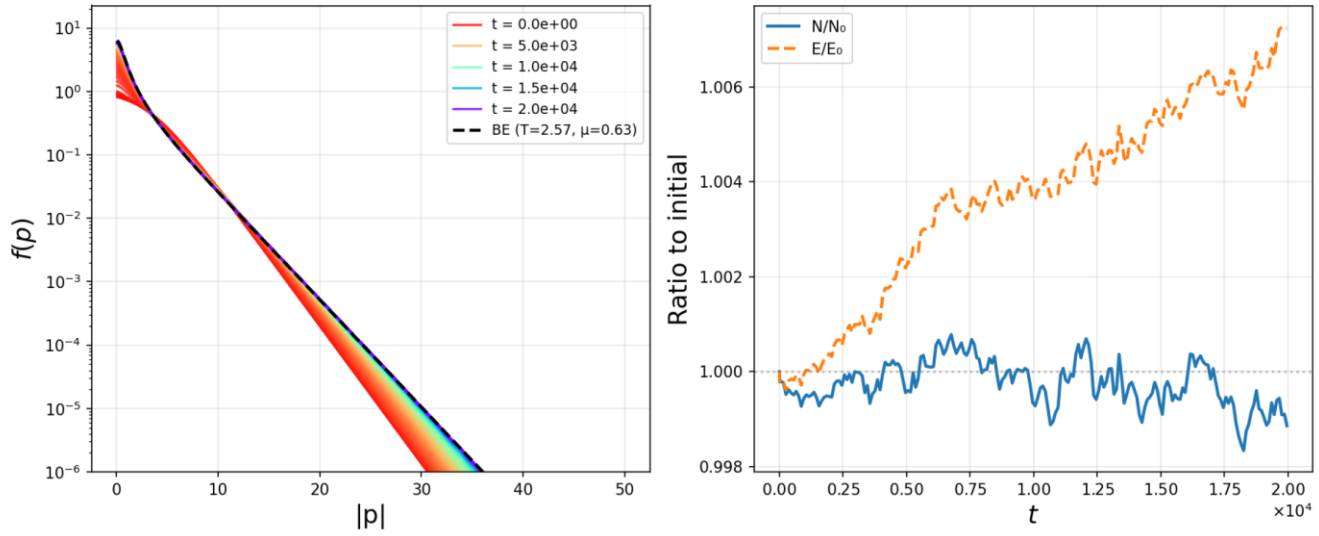




>v1.3

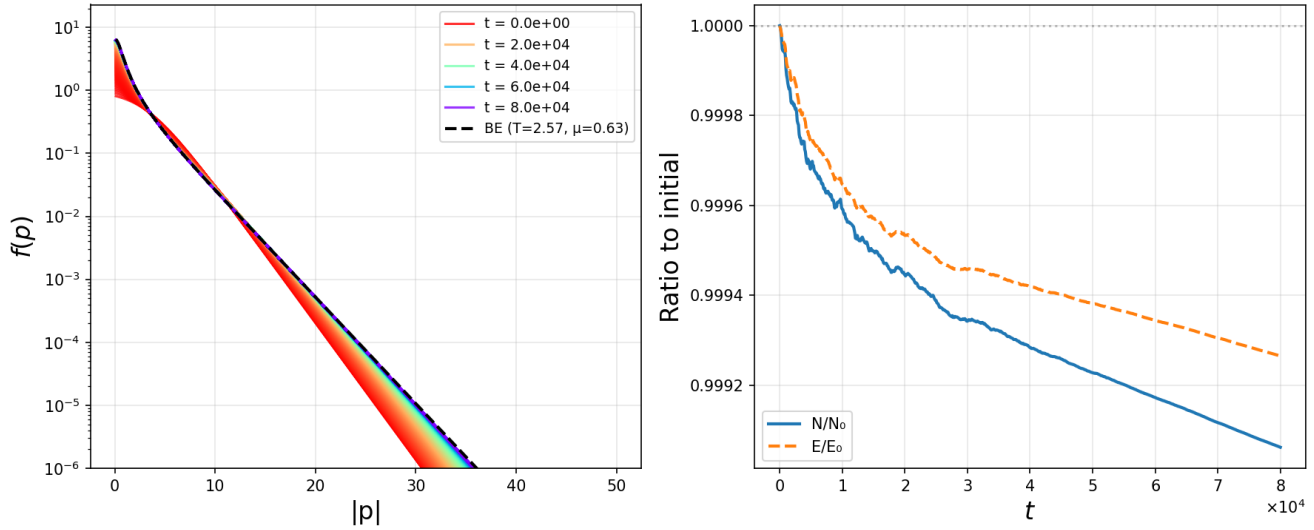
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Thermalization: 2 → 2 elastic scattering



- What evolves: f(p) itself, from a non-thermal start (left) — while N and E are tracked as the check (right).
- **The endpoint is a prediction, not a fit** — (T_eq, μ) fixed by the conserved N and E.
- No thermal input anywhere — the Bose–Einstein form emerges; N, E conserved to ~1% (MC noise); rate balance emerges between independently adapted integrators

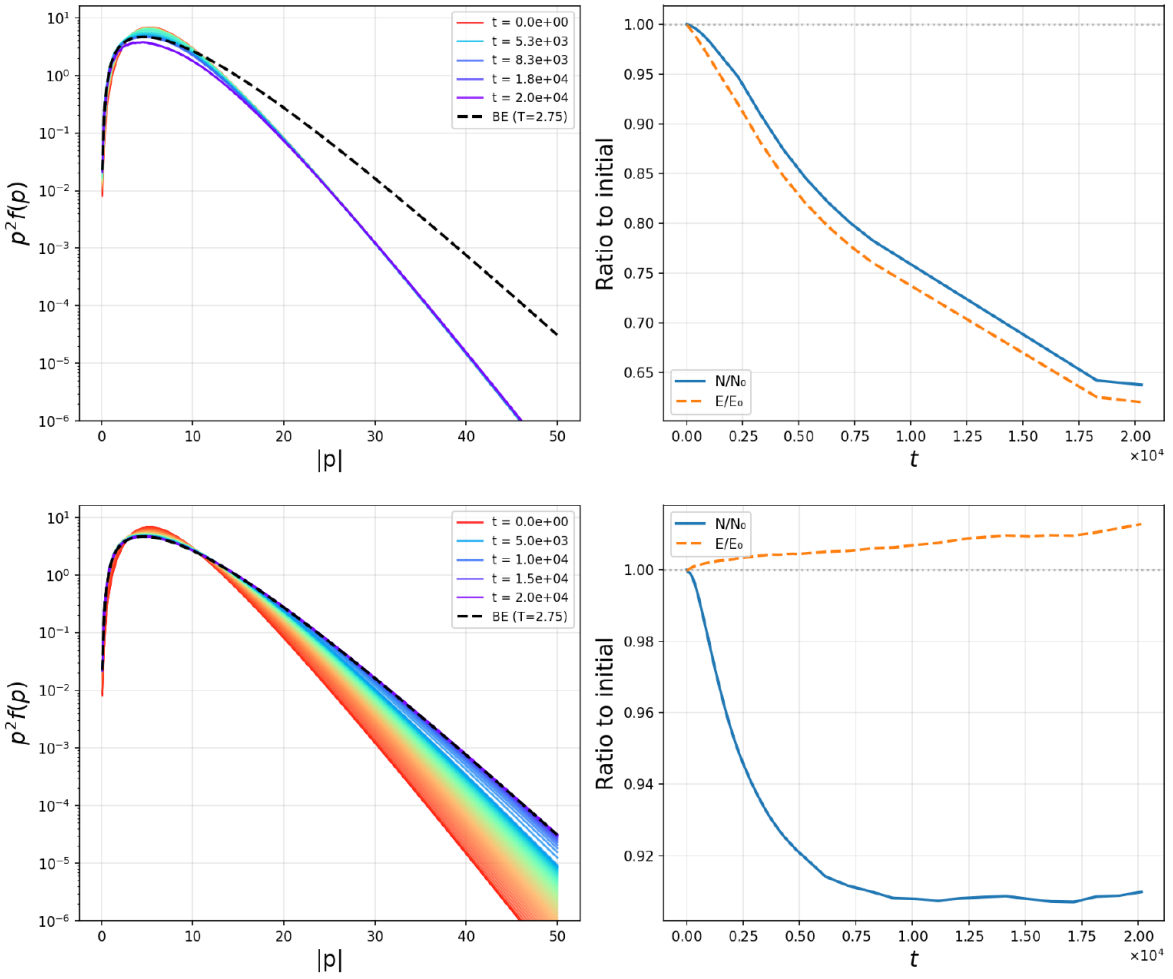
Thermalization: 2 → 2 elastic scattering



>v1.3

- What evolves: f(p) itself, from a non-thermal start (left) — while N and E are tracked as the check (right).
- **The endpoint is a prediction, not a fit** — (T_eq, μ) fixed by the conserved N and E.
- No thermal input anywhere — the Bose–Einstein form emerges; N, E conserved to **~0.1%**; gain and loss on the same samples

The key test: φφ ↔ φφφ



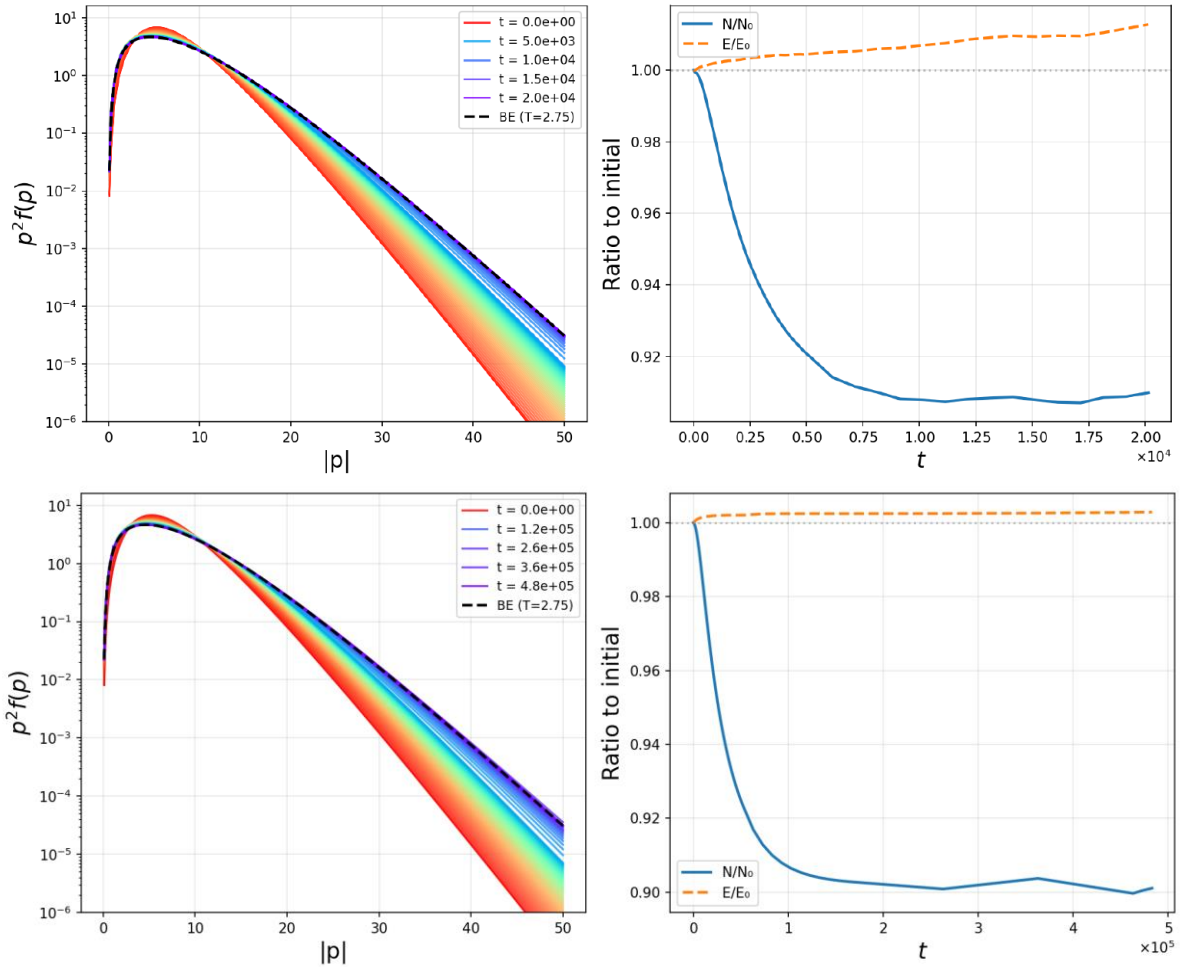
Top: naive implementation (C3 only)

- ~ 40% energy loss — systematic, not statistical
- wrong equilibrium: missing C2 distorts the production/absorption balance

Bottom: full 2 C2 + 3 C3

- energy conserved to ~ 1%
- f → Bose–Einstein with μ = 0 (chemical eq.: 3μ = 2μ)
- N/N0 → 0.9: the predicted ~10%
- net 3 → 2, μ → 0

The key test: φφ ↔ φφφ



Top: paper

- $|M|^2$ =1, no 1/(2! 3!) applied

Bottom: now (>v1.3)

- time axis ×12: $|M|^2$ is now bare ($\lambda_5 = 1$)

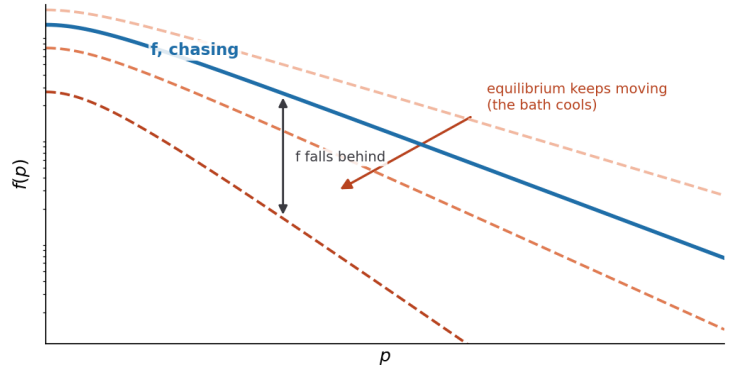
What changed since the paper

	paper (CPC)	now (v1.3)
dimensions	3n − 6 (2→2: 6, 2→3: 9)	3n − 8 (2→2: 4, 2→3: 7, 1→2: 1)
energy conservation	Gaussian δ of adapted width	exact two-body kinematics, every sample
gain & loss	two independent integrators	one joint sample, correlated net
what you write for a process	$\|M\|^2$ with symmetry factors and couplings folded in, plus a coupling argument	the bare $\|M\|^2$ of the Feynman rules — symmetry factors, multiplicities and degrees of freedom are the solver's
what you tune	the δ-width (delta_width, adapt_width) and error thresholds	nothing: no width, one neval
initial conditions	a function	a function, a table or a file; an empty species seeds itself
cost	Supercomputer, 272–544 ranks	laptop, 8 ranks

From flat spacetime to the expanding universe

- Part I was a closed box: the system reaches equilibrium and stays. Now the bath cools underneath:

  equilibrium itself becomes a moving target



- The interactions chase that target while H sets the pace of the cooling — when the chase is lost, f falls behind: freeze-out

- Numerically this regime is **stiff**: steps with H, not Γ — exponential integrators

Dark matter freeze-out

- When annihilation falls behind the expansion, the abundance stops tracking equilibrium and freezes — this is the relic

- This story is usually told with one number, n(t) — but the thing that actually freezes is the full distribution f(p, t)

- The difference matters when annihilation depends on where a particle sits in f

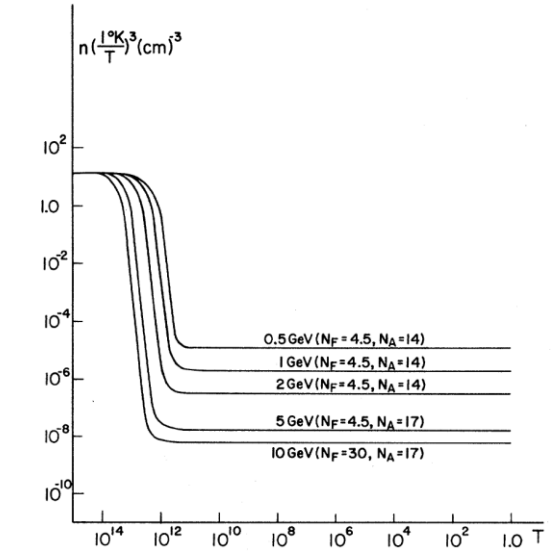


FIG. 1. n/T^3 vs T for a variety of special cases of m_L, N_F, and N_A.

Lee, Weinberg (1977)

Sub-threshold annihilation (aka forbidden DM)

$$\mathcal{L} \supset -\frac{\lambda}{4}\phi_1^2\phi_2^2 + y_f\phi_2\bar{f}f$$

(annihilation & elastic) (keeps ϕ_2 thermal)

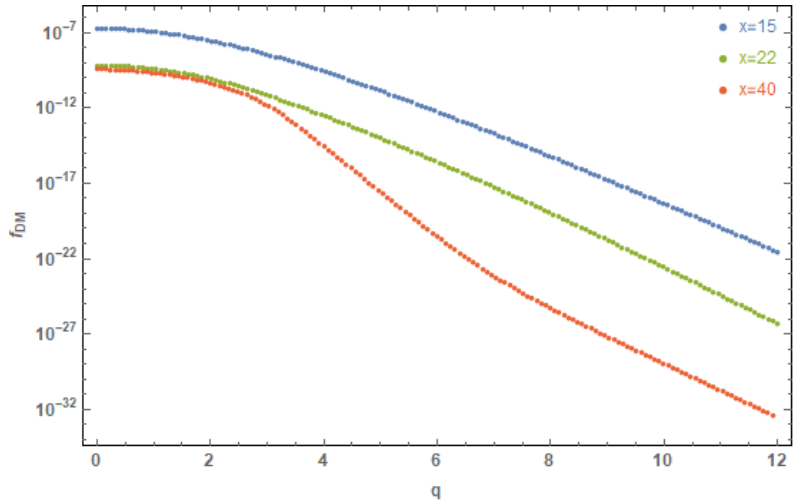
ϕ_1 = dark matter · ϕ_2 = bath, $m_2/m_1 = 1.1$ → sub-threshold

- As T drops, the typical ϕ_1 no longer has the energy to annihilate — only a window of f above threshold can
- Annihilation drains that window; scattering must refill it — the relic is set by that competition: by the shape of f

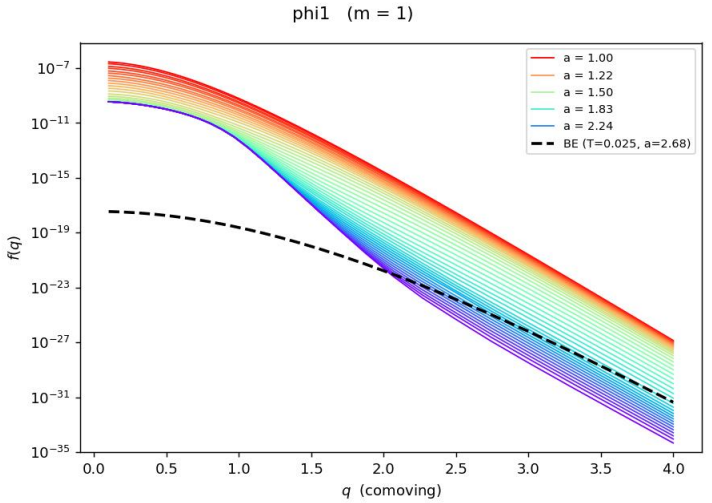
Dark matter relic abundance beyond kinetic equilibrium,
Binder, Bringmann, Gustafsson, Hryczuk (2021)

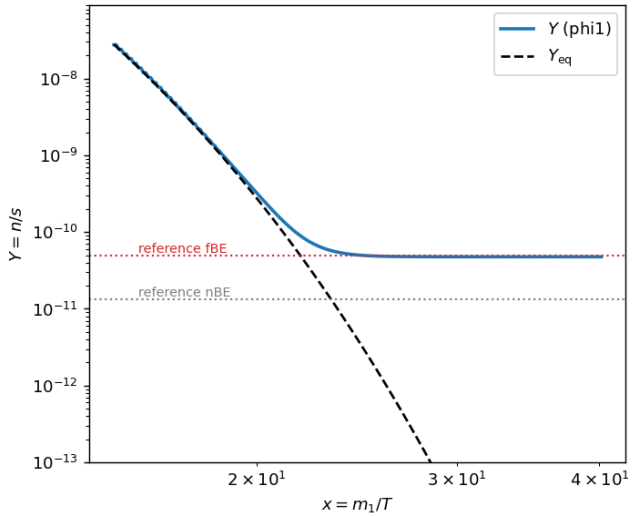
Sub-threshold annihilation: results

- Cross-checked against DRAKE



DRAKE

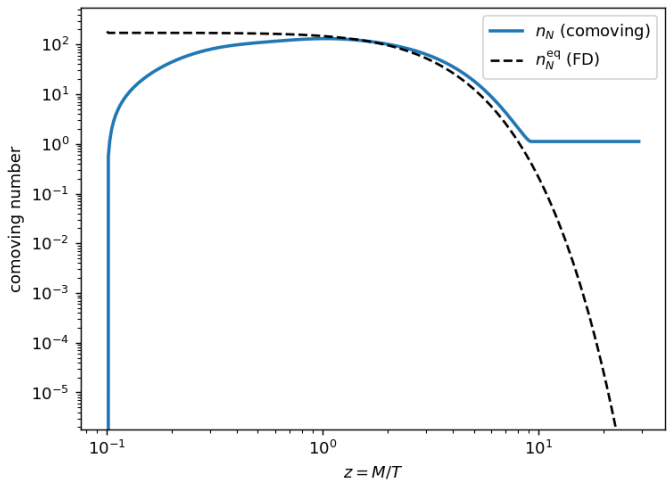




BEST

- Only particles above threshold can annihilate — the surviving low-energy bulk becomes the relic.
- Computed from the full f, that relic lands on the reference fBE line; the nBE misses it by ~×4.

Both at once: a stress test



One run, everything on:

- 4 processes — decays N ↔ Φ l, Φ ↔ l l · elastic Φ l ↔ Φ l, l l ↔ l l
- 3 species, all evolved — a Majorana fermion, a scalar, a lepton; Fermi–Dirac and Bose–Einstein; nothing prescribed
- exact kinematics through a closing threshold · time-dependent masses · expansion
- a stiff bath (Γ/H up to 10^4) on the full-Jacobian step · N starts empty and seeds itself

Outlook

- Custom collision terms
  - in-medium kernels beyond n → m: LPM splitting, soft scattering
  - the user writes the kernel; the solver steps it like any process — masses and rates can read f at every step
  - → hard decay products thermalizing in a plasma: reheating

- CP violation
  - forward and reverse $|M|^2$ separately; antiparticles as species
  - CP pairs on the same samples: the asymmetry carries only its own noise
  - → leptogenesis

Summary

So far

- Full Boltzmann equation, solved directly, any n → m
- Validated: 2 → 2 benchmark, 2 ↔ 3 thermalization
- Expanding universe: sub-threshold freeze-out, a stress test
- Since the paper: exact kinematics, more accurate, on a laptop

Ahead

- Dark-matter models: 3 → 2, freeze-in to freeze-out, dark sector thermalization
- Reheating: Standard Model thermalization
- Leptogenesis: CP violation

github.com/best-hep/best All the BEST.

The code in one slide

```
import numpy as np
from besthep import BEST

lam5 = 1.0                # L = -λ5 φ^5/5!

def M2(momenta):          # bare |M|^2 of the Feynman rule
    return np.full(momenta.shape[2], lam5**2)

solver = BEST(q_min=0.1, q_max=20.0,
              n_grid=64)
solver.initialize_species(
    'phi', init_f,
    stat='boson', mass=1.0)
solver.add_process(
    'cannibal',
    ['phi','phi'],
    ['phi','phi','phi'],
    M2)                   # no coupling=, no 1/(2!3!)
for step in range(n_steps):
    solver.evolve_step(dt=dt)
```

- One process, one line: dimensions, phase space and the identical-particle decomposition follow from the declaration
- The bare |M|² of the Feynman rules: symmetry factors, multiplicities and degrees of freedom are the solver's
- Nothing to tune: no δ-width, one neval
- Fermions and bosons · time-dependent masses · expansion · exponential steppers for stiff problems
- Python only (numpy, scipy, mpi4py, vegas) · one pickle = full state, resume anywhere

Energy conservation requires the complete sum

Integrate E Ca(p) over all momenta using the full decomposition:

$$\int \frac{d^3p}{(2\pi)^3}\, E\, C_a(p) \propto \int d\Pi_{\text{all}} \left(\sum_{i \in \text{in}} E_i - \sum_{j \in \text{out}} E_j \right) |\mathcal{M}|^2\, \Lambda$$

vanishes by the energy–momentum δ-function — but only if the sum over all positions is complete.

- omit C2 for 2 ↔ 3: $\int E\, C_3\, p^2\, dp \neq 0 \Rightarrow$ systematic energy non-conservation (not an MC artifact)
- invisible in the nBE: momentum integration absorbs the sum over positions
- invisible to all previous fBE solvers: for 2 → 2, Cnα = Cnβ automatically

The rule, forced by energy conservation: C_a = n_α C_α + n_β C_β — every placement, weighted by its slots.